MA21 : Macroscopic characterisation for material science

Tutorial 1: UV-VIS optical spectroscopy

- Units I:

a- Find the formulae to convert nm (λ) into cm^{-1} (σ) and in eV (E).
b- Give the spectral range for visible in eV, knowing that the eye is sensitive to the wavelength from 400 to 750 nm.
c- Give the expression of Δλ as a function of ΔE, comments...

- Units II:

$Al_{x=0.3}Ga_{1-x=0.7}As$ is a semiconductor alloy made of 3 elements (Al, Ga and As) with variable Al and corresponding Ga concentrations. This III-V semiconductor is commonly used as an optoelectronic material for Light emitting devices. By combining different III-V elements together one can tune the emission wavelength from the UV to the NIR (cf. the graph below).

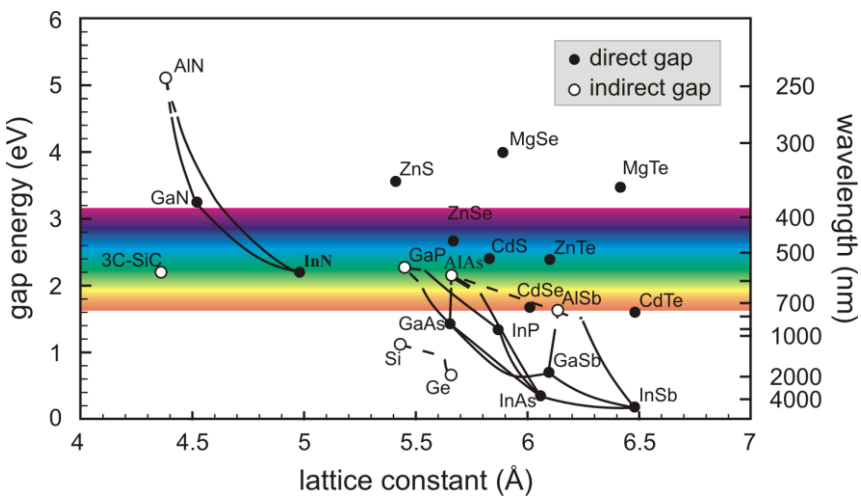

Figure 1 : Bandgap energy of major III-V and II-VI semiconductors as a function of the lattice constant. Only the materials of similar lattice constant can be combined together (as an example $Al_xGa_{1-x}As$ is grown on a GaAs substrate). II, III, V, VI refer to the column in the element periodic table.

a- Give the bandgap energy in eV and in nm using the formula below:

$x<0.45$ $E(x) = 1.424+1.247x$ (eV)
$x>0.45$ $E(x) = 1.9+0.125x+0.143x^2$ (eV)

b- What is the maximum of the energy bandgap change for x ε [0, 0.45]?

c- Propose a simple spectroscopic method to measure the bandgap.

Given data:

$h = 6.64\ 10^{-34}$ J.s

$c = 3\ 10^{8}$ m/s

$1eV = 1.6\ 10^{-19}$ J

-Absorption I :

a- With the help of the expression for electromagnetic waves (linear polarization along z and propagation direction along x) find the Beer-Lambert law. We will introduce the complex refractive index n_c.

We recall the hypothesis for the Beer –Lambert law : the intensity I of a monochromatic radiation is reduced by dI (negative) after passing through a thickness dx of the material, with an absorption coefficient α (for a given wavelength), meaning:

$-dI/dx=\alpha I(x)$

We neglect here the interface effects due to the finite size of the sample.

b- Show that with an additional assumption, in the case where a compound is dissolved in an transparent (non-absorbing) medium of thickness d, a « simple » absorbance measurement allows to measure the molar concentration, c. Find a relation between A, the absorbance, d, c and another characteristic called ε_λ (molar absorption coefficient).

Discuss the resulting units of this coefficient.

c- What happens if we mix several compounds? Find a method to measure the contribution of the solvent of the solution to be analysed.

d- Example of calculation:

An aqueous solution of permanganate of potassium ($c=1.28\ 10^{-4}$ mol.L^{-1} has a transmittance of 0.5 at 525 nm if we use a sample holder of 10 mm of optical path).

i- Find the molar absorption coefficient of permanganate for this wavelength.

ii- If we double the concentration, calculate the absorbance and the transmittance of the new solution?

-Absorption II: Units

As just seen, chemists use the molar absorption coefficient, ε_λ (L.mol^{-1}.cm^{-1}). Physicists rather talk about the absorption cross section, σ (cm^2). Do they talk about the same thing?

We recall that for a chemist, transmission is $T=10^{-DO}$ with $DO=\varepsilon_\lambda dc$. For a physicist $T=e^{-\alpha d}$ with $\alpha=\sigma N$ and N being the number of atoms (molecules) per cm^{-3}.

-Absorption III: Effect of the interfaces

a- What happens for the expression of the absorbance when we take into account the interfaces?
Define the reflectance R.

b- What happens to the experimental protocol for measuring correctly the absorbance in this case?

c-What happens in this case of the light is also diffused by the sample?

d- What happens if we take the coherence of the laser into account?